The kernel report

(Kernel Recipes 2026 edition)

Jonathan Corbet
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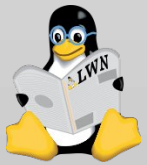
Kernel Recipes 15th



The Kernel Report

Jonathan Corbet

hupstream



Recent release history

Release	Date	Commits	Developers
6.17	Sep 28	13,089	2,038
6.18	Nov 30	13,710	2,134
6.19	Feb 8	14,344	2,137
7.0	Apr 12	14,251	2,361
7.1	Jun 14	15,849	2,478
7.2	Aug 16	16,418	2,652
7.3	(Oct 18)	17,056	2,720



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BORING!



The constants of kernel development

Commit counts go up
Developer counts go up



The constants of kernel development

A kernel release takes 9 or 10 weeks

Exceptions:

- 4.15 January 2018 (11 weeks)
- 3.1 October 2011 (12 weeks)



The constants of kernel development

Lots of new stuff is added

- The folio transition
- sched-ext (sub-schedulers)
- Rust is no longer experimental
- Live update orchestrator
- Time-slice extension
- BPF in io_uring
- Swap reimplementation
- Hundreds of device drivers

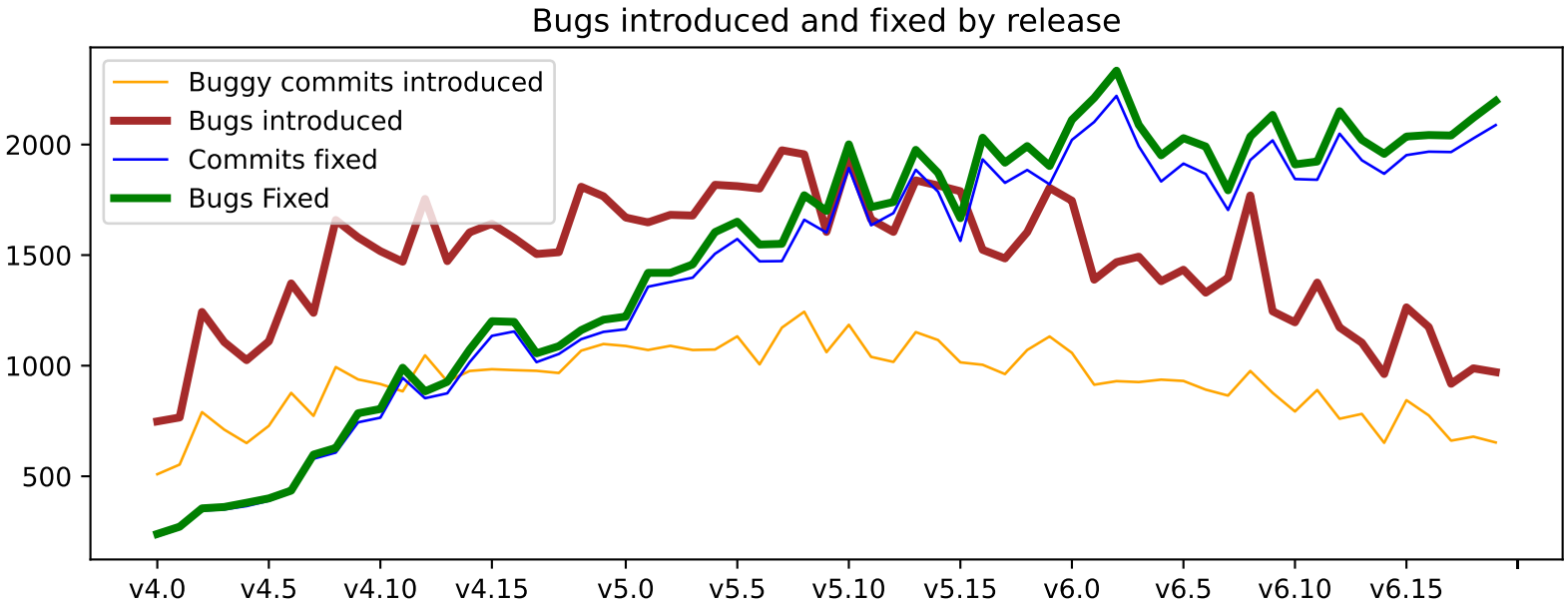


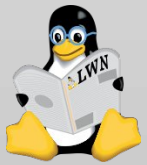
Business as usual

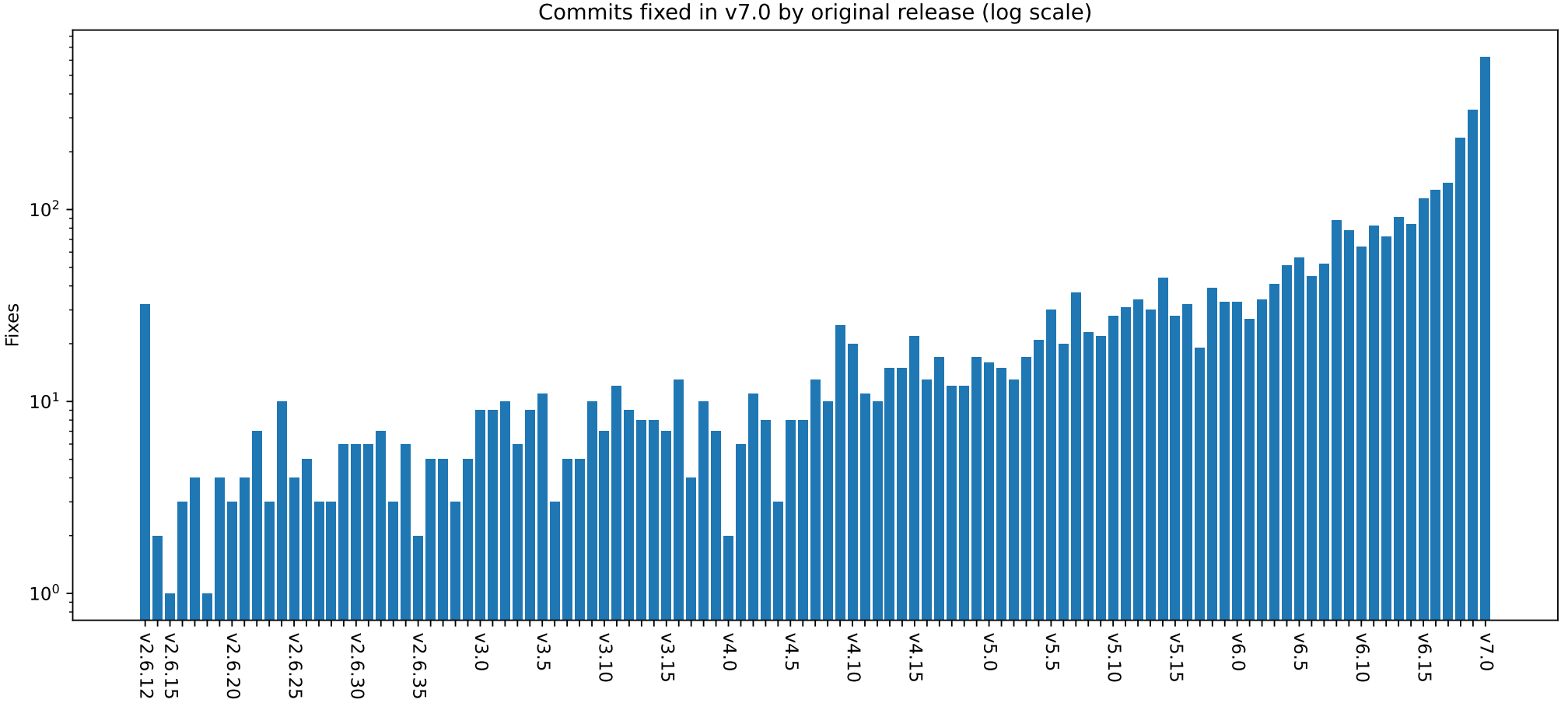


Business as usual?

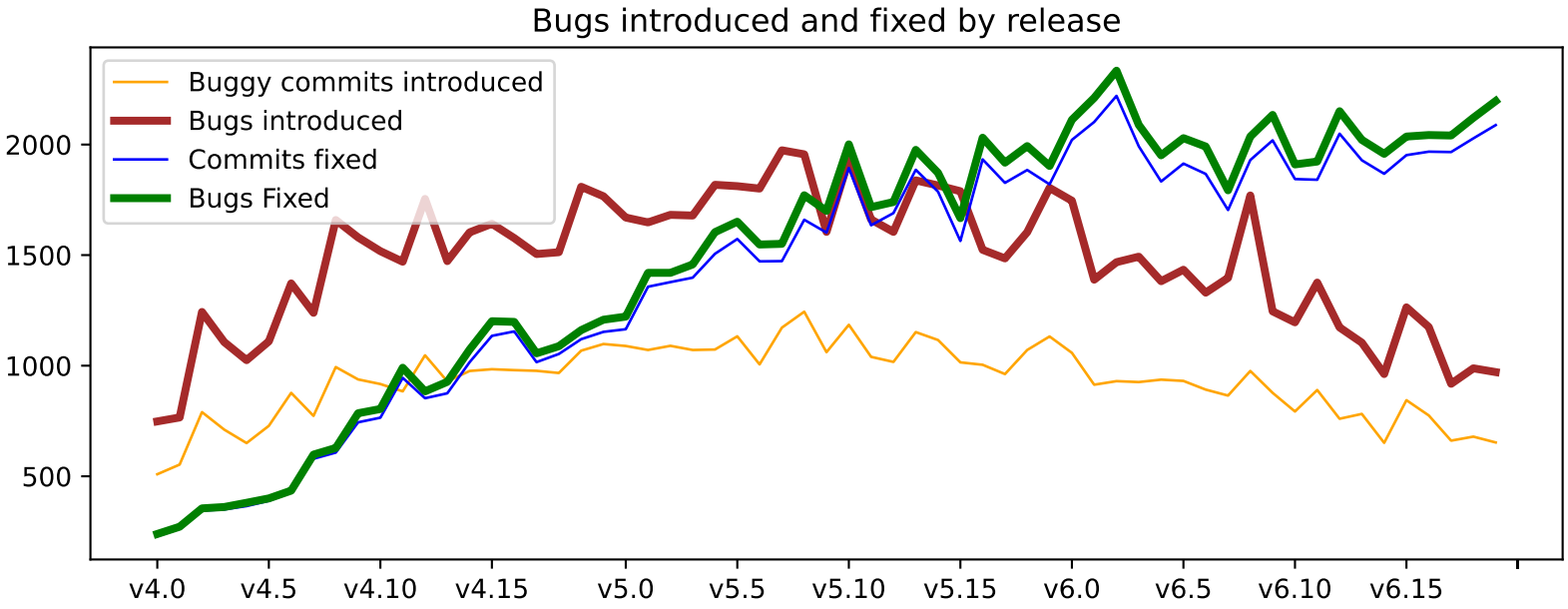


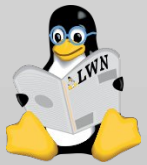

Bugs introduced and fixed by release
Buggy commits introduced
Bugs introduced
Commits fixed
Bugs Fixed
2000
1500
1000
500
v4.0
v4.5
v4.10
v4.15
v5.0
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v6.5
v6.10
v6.15

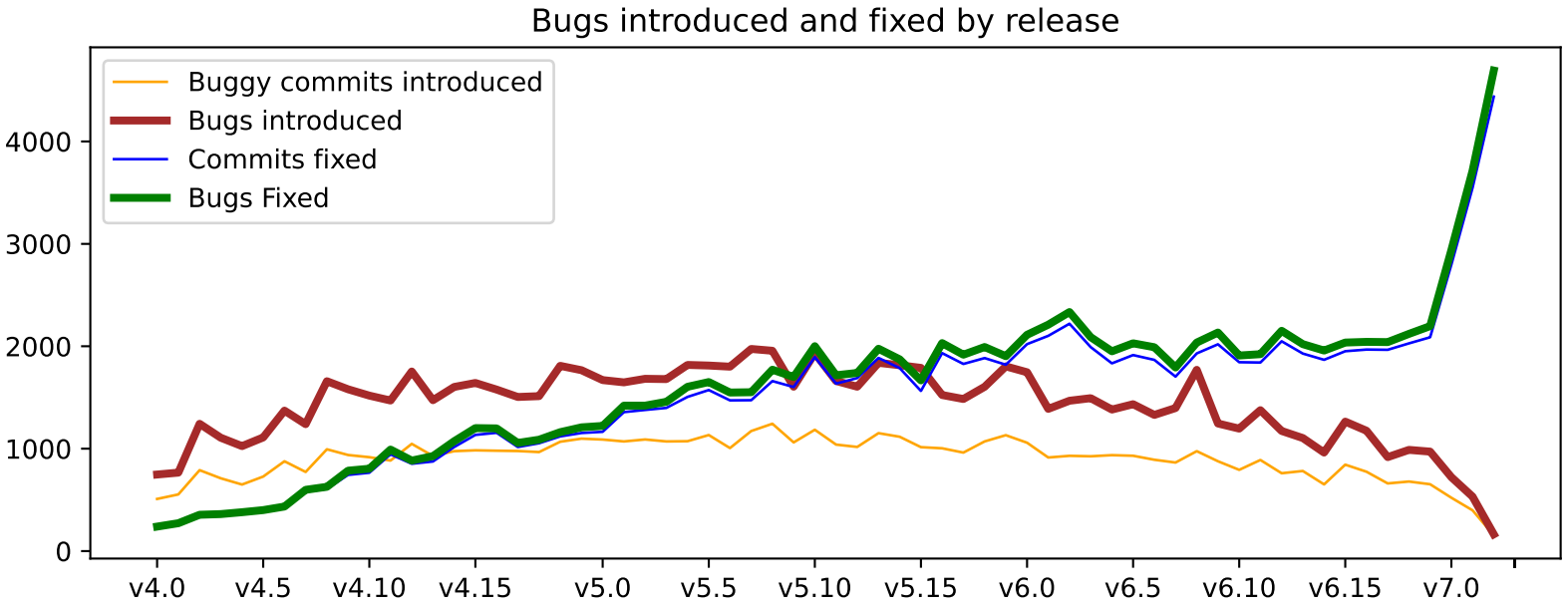

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Commits fixed in v7.0 by original release (log scale)
Fixes
10^2
10^1
10^0
v2.6.12
v2.6.15
v2.6.20
v2.6.25
v2.6.30
v2.6.35
v3.0
v3.5
v3.10
v3.15
v4.0
v4.5
v4.10
v4.15
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v6.0
v6.5
v6.10
v6.15
v7.0

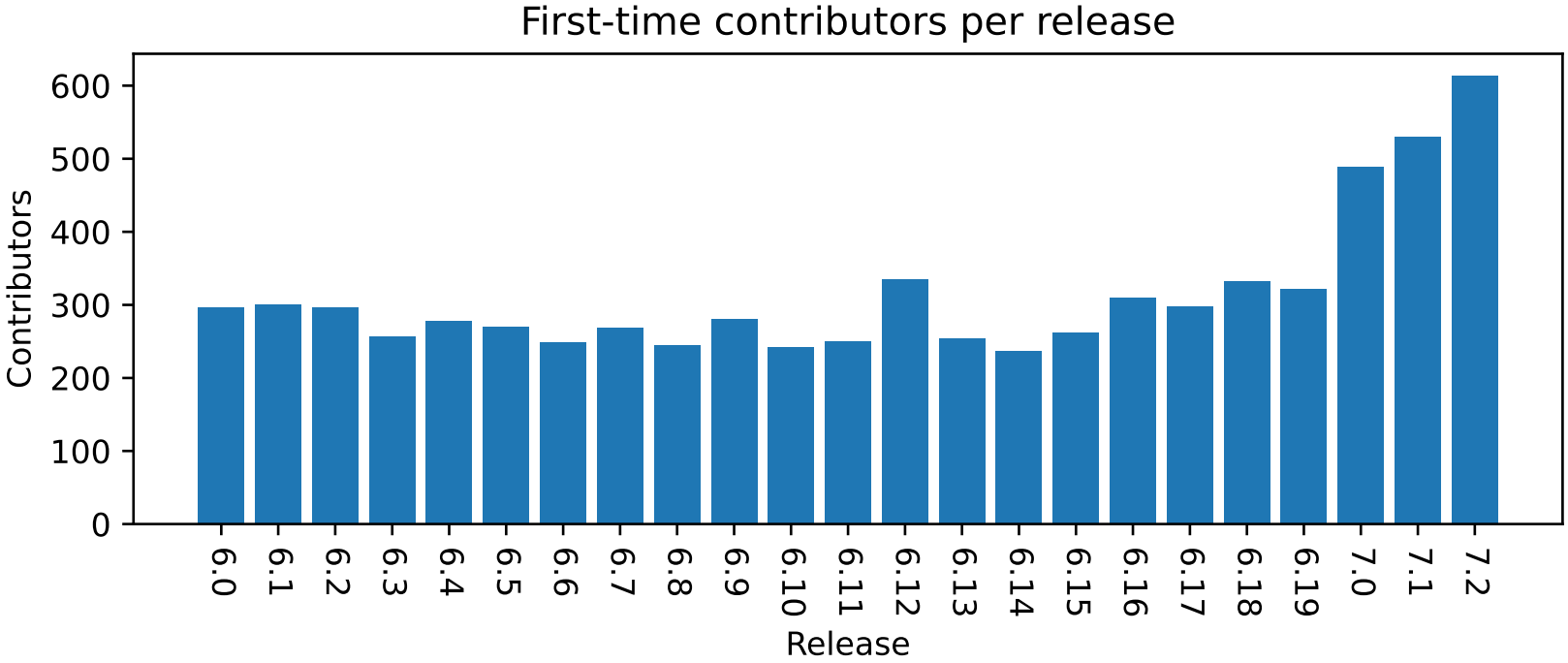

LWN


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v6.0
v6.5
v6.10
v6.15


LWN


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4000
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0
v4.0
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v5.0
v5.5
v5.10
v5.15
v6.0
v6.5
v6.10
v6.15
v7.0


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First-time contributors per release
Contributors
0
100
200
300
400
500
600
6.0
6.1
6.2
6.3
6.4
6.5
6.6
6.7
6.8
6.9
6.10
6.11
6.12
6.13
6.14
6.15
6.16
6.17
6.18
6.19
7.0
7.1
7.2
Release


LWN

Large language models



LLMs in kernel development

Patch review (sashiko)



Patch authors are expected to proactively look into the AI-generated reviews and handle such feedback as any other kind of review: either debate it or address it. In both cases a reply on the mailing list is expected.

Authors are strongly encouraged to run LLM reviews on the posted patches in advance of the actual post. Large series triggering a significant amount of AI-generated feedback will likely get little attention from maintainers and reviewers.

— Networking subsystem policy



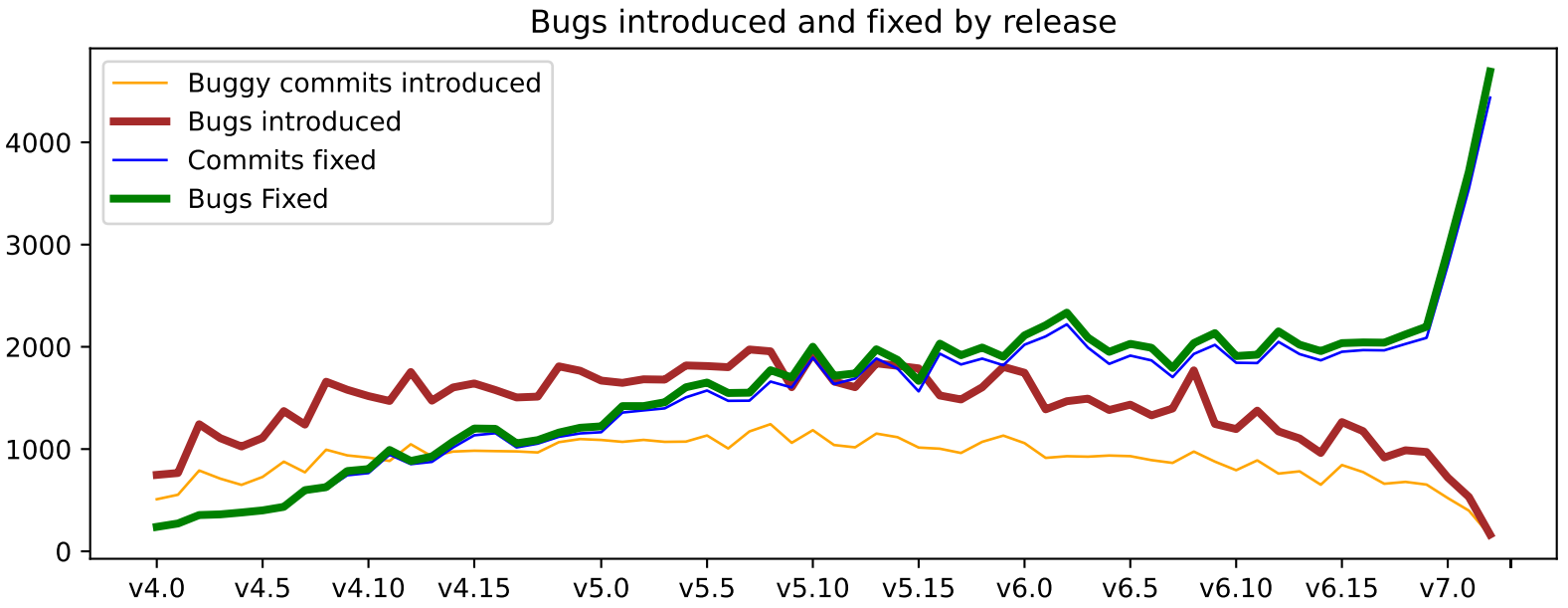
LLMs in kernel development

Finding bugs



LLMs in kernel development

Finding bugs





Security bugs too...

> The code was public, but it was complicated enough, and there was so much of it, that finding a serious bug took rare skills and a lot of time. AI removes that barrier. What's left is how fast we triage, fix, and ship.
> — Sasha Levin



LLMs in kernel development

Maintainer tasks?

> I expect our direction for the next release will be to tweak the reviews a little bit more, but start shifting focus to letting the LLMs take care of the busy work - managing patchwork, automating common process complaints, editing commit messages, and maybe applying patches which already got "reviewed-by" tags from people we trust...
>
> — Jakub Kicinski



LLMs in kernel development

Creating patches



> Changelogs from Asian developers have become a lot easier to read in recent times.
> — Andrew Morton



An LLM was used to first determine where the bottlenecks were then to figure out how to improve them.

It generated a lot of code, much of it hideous.
— Lorenzo Stoakes speeds up kernel builds



Summary nntp+archive.lwn.net:lwn.kernel.lkml

File Options Buffers Tools Gnus Post Threads Article Score Agent Help

```
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U:@---  archive:l.k.lkml [3749477]     9%   L53    (Summary Plugged)
```



Assisted-by tags per release

Release	#tags	
6.18	1	
6.19	2	
7.0	31	
7.1	301	
7.2	1111	
7.3	1606	(-rc4)



Some questions

Who understands all of this code?



Who understands this code?

Subsystems touched by one first-time 7.2 contributor:

input	nvme	F2FS	AppArmor
exfat	NTFS	Landlock	mac80211
partitions	UDF	isofs	virtio
libata	net/sched	orangefs	IIO
SCSI	ceph	selinux	thunderbolt



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Some questions

Who understands all of this code?
Who is reviewing all of this code?



Some questions

Who understands all of this code?
Who is reviewing all of this code?
Who stands behind this code?



Other concerns

Access to LLM-based tools



Other concerns

Access to LLM-based tools

Rugpulls



Other concerns

Access to LLM-based tools
Rugpulls
Copyright worries



Other concerns

Access to LLM-based tools
Rugpulls
Copyright worries
Ethical and environmental concerns



LLM use appears set to keep growing



> Is there really no acceptable option beside forcing everybody to swallow the generative AI pill ? No option for people with ethics concern to be still considered part of the kernel community ? No option but complete surrender and pledging allegiance ? This is how I increasingly feel being treated, and I know I'm not alone.
>
> — Laurent Pinchart



What else?



Generational change



Andrew Morton announces his (gradual) retirement









Losing long-time developers

Why?

- Changes in the community
- We’re not getting any younger
- Bad things happen



Steve French, Dan Williams







Preparing for change

How do we...

- Preserve the experience/knowledge?



Preparing for change

How do we...

- Preserve the experience/knowledge?
- Bring up new maintainers?

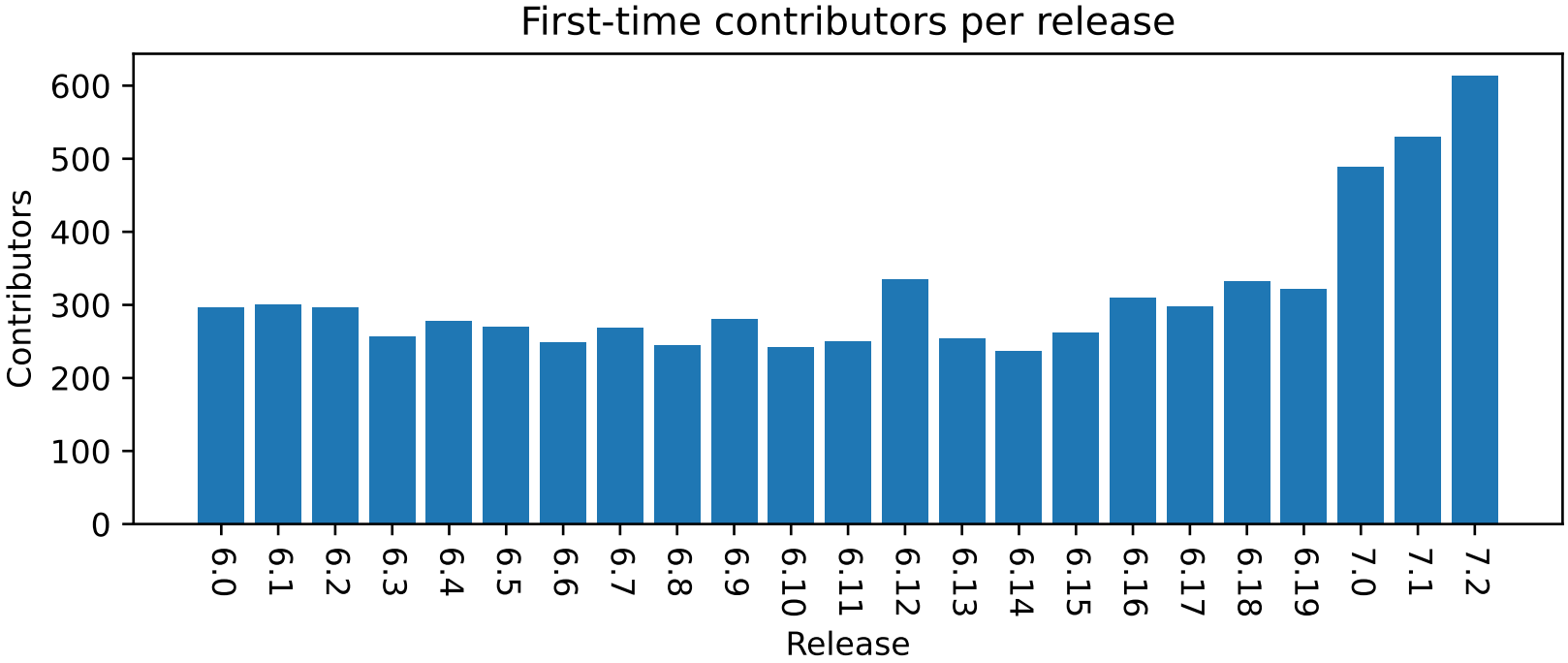


Preparing for change

How do we...

- Preserve the experience/knowledge?
- Bring up new maintainers?
- Continue to attract developers?




First-time contributors per release
Contributors
0
100
200
300
400
500
600
6.0
6.1
6.2
6.3
6.4
6.5
6.6
6.7
6.8
6.9
6.10
6.11
6.12
6.13
6.14
6.15
6.16
6.17
6.18
6.19
7.0
7.1
7.2
Release


LWN

Preparing for change

How do we...

- Preserve the experience/knowledge?
- Bring up new maintainers?
- Continue to attract **long-term** developers?



Rust



BPF



BPF developments

io_uring integration
sched_ext sub-schedulers
blk-iocost integration
BPF-driven memory reclaim
...



Development tools



Interesting tools news

(beyond LLMs...)

Self tests

b4



Events



Thank you!

